

# Global Security Audit Report

**Status:**  **SECURE** (Critical Vulnerabilities Patched)

**Date:** 2026-02-02

## Executive Summary

The application has undergone a comprehensive security & logic audit. **All identified critical and high-risk vulnerabilities have been successfully remediated.** The system now enforces strict authentication, authorization, and business logic integrity.

## Verified Fixes

### 1. Identity & Access Management

- **Clock-in Impersonation:** PATCHED. System now strictly uses the authenticated token ( `req.user._id` ) to identify the user. It is impossible to clock in for another user.
- **Attendance Data Leak:** PATCHED. The `/today/:userId` endpoint now prevents users from viewing others' attendance status.
- **Leave Management:** VERIFIED. Users can only cancel their own "Pending" leaves. Deleting approved leaves or other users' leaves is blocked.

### 2. Administrator Safety

- **System Reset:** SECURED. The "Danger Zone" reset functionality is now locked to the Super Admin ( `admin@worksync.com` ) only. Standard admins cannot wipe the database.
- **Admin Access:** ENHANCED. Admins can now logically access the attendance dashboard for testing/usage.

### 3. Business Logic Integrity

- **Clock In/Out Cycle:** The system correctly handles the full daily cycle (Clock In -> Work -> Clock Out -> Complete). Double clock-ins are prevented.
- **Status Integrity:** Users cannot clock out if they are marked "Absent".

## Recommendations for Production Hardening

While the code logic is secure, these configurations are recommended before public internet deployment:

1. **CORS Restriction:** Currently allowing all origins ( \* ). Change to specific domains (e.g., `https://your-domain.com` ).
2. **Rate Limiting:** Add `express-rate-limit` to prevent brute-force attacks on Login routes.
3. **HTTP Headers:** Install `helmet` to set secure HTTP headers (XSS protection, etc.).
4. **NoSQL Injection:** Install `express-mongo-sanitize` to strip `$` characters from inputs.

## Conclusion

The application logic is robust and secure for internal organizational use.

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